Lecture 9: Hypothesis Testing

p-Values · Power Analysis · Permutation Tests · Multiple Testing

Previously, on Lecture 8...

Confidence interval: A random interval $[L, U]$ with $P(L \leq \theta \leq U) = 1 - \alpha$. The *procedure* is 95%.

Wald CI: $\hat{\theta} \pm z_{\alpha/2} \cdot \text{SE}$. Works when CLT kicks in. Use Wilson for proportions.

t**-interval:** Use when σ unknown and n small. Heavier tails $\Rightarrow$ wider CI.

Bootstrap: Resample with replacement $\Rightarrow$ SE and CI without formulas. BCa is the gold standard.

Two paths: Analytical when formulas exist; bootstrap when they don't. Same goal.

Today: We can estimate and quantify uncertainty. Now: is the effect **real** or **noise**?

Part I: The Logic of Testing

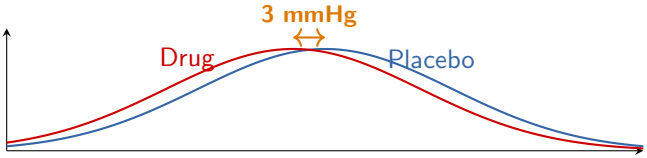
Is 3 mmHg a real effect or just noise?

The Drug Trial

Scenario: A pharmaceutical company tests a new blood pressure drug.

- $n = 100$ patients per group (drug vs. placebo)
- Placebo group mean: $\bar{X}_P = 140$ mmHg
- Drug group mean: $\bar{X}_D = 137$ mmHg
- Pooled SD: $S = 12$ mmHg

Question: Is this 3 mmHg reduction *real*, or could random chance produce it?



Innocent Until Proven Guilty

Courtroom

1. **Assume** defendant is innocent
2. **Examine** the evidence
3. **Convict** only if evidence is overwhelming

"Innocent until proven guilty"

Verdict: guilty / *not proven guilty*
(never "proven innocent")

Statistics

1. **Assume** H_0: no effect
2. **Compute** how surprising the data is
3. **Reject** H_0 only if the data is extremely unlikely under H_0

"No effect until proven otherwise"

Verdict: reject H_0 / *fail to reject* H_0
(never "H_0 is true")

H_0 and H_1

H_0 (null hypothesis)

"Nothing is happening"

The default / boring explanation

Examples:
Drug has no effect: $\mu_D = \mu_P$
The coin is fair: $p = 0.5$
Treatment A = Treatment B

H_1 (alternative)

"Something is happening"

What we want evidence *for*

One-sided: $\mu_D < \mu_P$
(drug lowers BP)
Two-sided: $\mu_D \neq \mu_P$
(drug changes BP)

Key: We never "prove" H_0. We either **reject** it (strong evidence against) or **fail to reject** it (insufficient evidence). Absence of evidence $\neq$ evidence of absence.

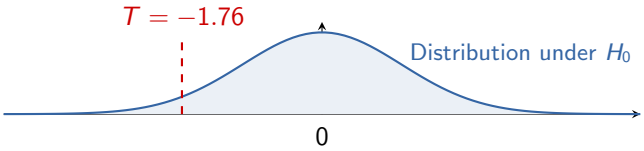
The Test Statistic

A **test statistic** T compresses the data into one number measuring "distance from H_0".

$$T = \frac{\bar{X}_D - \bar{X}_P}{\text{SE}} = \frac{\bar{X}_D - \bar{X}_P}{S\sqrt{2/n}}$$

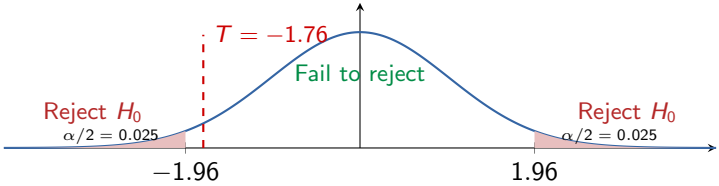
$$\text{Drug trial:} \quad T = \frac{137 - 140}{12\sqrt{2/100}} = \frac{-3}{1.70} = -1.76$$

Under H_0: $T \sim N(0, 1)$ approximately. Large $|T|$ = data far from H_0 = evidence against H_0.



The Rejection Region

Choose a **significance level** α (typically 0.05) *before* looking at the data.



Drug trial: $|T| = 1.76 < 1.96 \Rightarrow$ **fail to reject** H_0 at $\alpha = 0.05$.

The 3 mmHg reduction is *not statistically significant* at the 5% level.

Part II: Two Ways to Be Wrong

False alarms and missed discoveries

Type I and Type II Errors

	Fail to reject H_0	**Reject H_0**
H_0 true	**Correct** True negative $P = 1 - \alpha$	**Type I Error** False alarm $P = \alpha$
H_0 false	**Type II Error** Missed detection $P = \beta$	**Correct** Power $= 1 - \beta$

Type I and Type II Errors

	Fail to reject H_0	**Reject H_0**
H_0 true	**Correct** True negative $P = 1 - \alpha$	**Type I Error** False alarm $P = \alpha$
H_0 false	**Type II Error** Missed detection $P = \beta$	**Correct** Power $= 1 - \beta$

Type I (false positive):
Telling a man he's pregnant.
Convict an innocent person.
Approve a useless drug.

Type II (false negative):
Telling a pregnant woman she's not.
Free a guilty person.
Miss an effective drug.

Part III: The p-Value

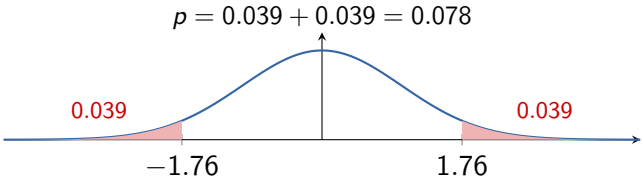
The most used — and most abused — number in science

The p-Value: Definition

The **p-value** is the probability of observing data *this extreme or more*,

assuming H_0 is true:

$$p = P\big(|T| \geq |T_{\text{obs}}| \;\big|\; H_0\big)$$



Drug trial: $p = 0.078 > 0.05 = \alpha \;\Rightarrow\;$ fail to reject H_0.

Rule: Reject H_0 if and only if $p \leq \alpha$.

What the p-Value Is NOT

✗ "$p = 0.03$ means there's a 3% chance H_0 is true."

Reality: $p = P(\text{data} \mid H_0)$, not $P(H_0 \mid \text{data})$. The transpose fallacy!

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Reality: p says nothing about $P(H_1)$. For that you need Bayes.

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✗ "Smaller p = bigger effect."
Reality: p depends on *both* effect size and n. With $n = 10^6$, any tiny effect gives $p \approx 0$.

Statistical vs Practical Significance

With enough data, *any* nonzero effect becomes "statistically significant."

Scenario A

$n = 100{,}000$ per group
Drug lowers BP by **0.1 mmHg**
$p < 0.001$

Statistically significant
Clinically meaningless

Would you take a pill every day for 0.1 mmHg?

Scenario B

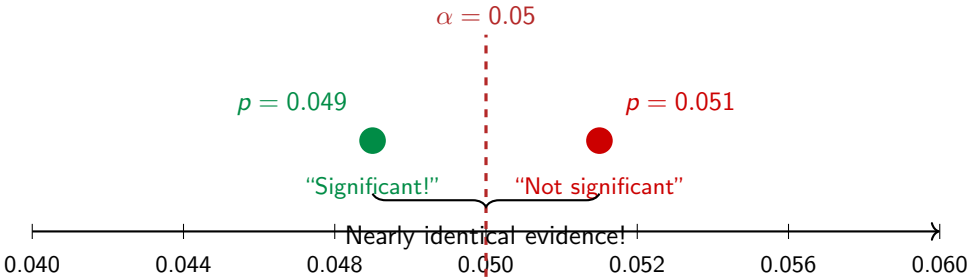
$n = 15$ per group
Drug lowers BP by **8 mmHg**
$p = 0.12$

Not significant
Clinically important

Study was underpowered, not the drug's fault.

Always report: **(1) effect size**, **(2) confidence interval**, **(3) p-value**.
Ask "**How big** is the effect?", not just "Is it nonzero?"

$p = 0.049$ vs $p = 0.051$



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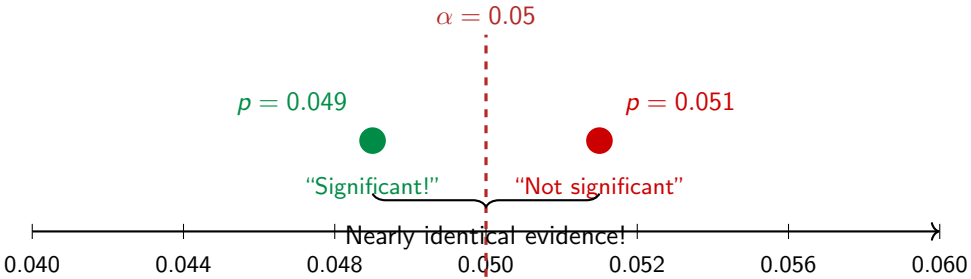


The 0.05 threshold is a **convention**, not a law of nature.

ASA (2016): "Scientific conclusions should not be based only on whether a p-value passes a specific threshold."

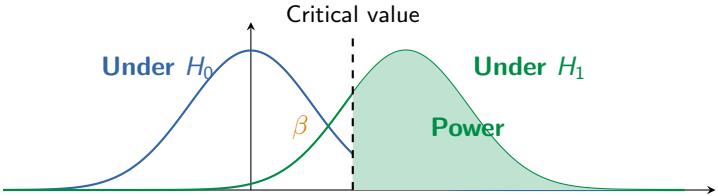
Part IV: Power Analysis

How to design a study that can actually find what you're looking for

Power: The Ability to Detect Real Effects

$$\text{Power} = P\big(\text{reject } H_0 \;\big|\; H_1 \text{ is true}\big) = 1 - \beta$$

Convention: aim for power $\geq$ 0.80 (80%).



What Determines Power?

1. Effect size $\delta \uparrow$

Bigger real effect
$\Rightarrow$ easier to detect
$\Rightarrow$ higher power

2. Sample size $n \uparrow$

More data $\Rightarrow$ narrower SE
$\Rightarrow$ distributions separate
$\Rightarrow$ higher power

3. Significance level $\alpha \uparrow$

Larger $\alpha \Rightarrow$ bigger
rejection region
$\Rightarrow$ higher power (but more Type I)

4. Variability $\sigma \downarrow$

Less noise $\Rightarrow$ clearer signal
$\Rightarrow$ narrower distributions
$\Rightarrow$ higher power

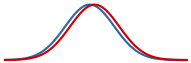
You control n and α. Nature controls δ and σ.
Power analysis = choosing n so that you can detect a meaningful δ.

Effect Size: Cohen's d

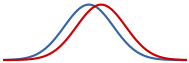
$$d = \frac{\mu_1 - \mu_2}{\sigma}$$

Effect in units of standard deviations.

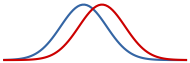
$d = 0.2$ (small)



$d = 0.5$ (medium)



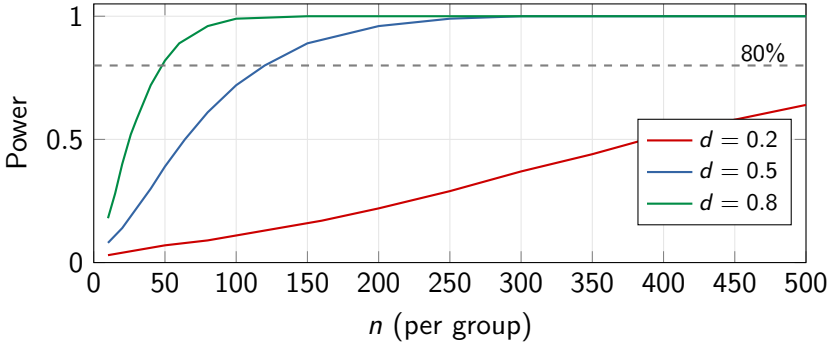
$d = 0.8$ (large)



Drug trial: $d = 3/12 = 0.25$ (small).
Effect size is **independent of sample size** — it measures the *phenomenon*, not the study.

Power Curves: Planning Your Study

Two-sample z-test, $\alpha = 0.05$, two-sided



Required n per group for 80% power: $d = 0.8$: $n \approx 26$ $d = 0.5$: $n \approx 64$ $d = 0.2$: $n \approx 394$

Sample Size Formula

$$n_{\text{per group}} = \frac{2\left(z_{\alpha/2} + z_{\beta}\right)^2}{d^2}$$

where $d = \delta/\sigma$ and $z_{\beta} = 0.84$ for 80% power.

Large effect
$d = 0.8$
$n = 26$

Medium effect
$d = 0.5$
$n = 64$

Small effect
$d = 0.2$
$n = 394$

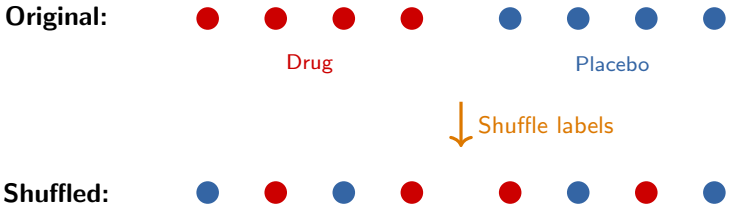
An underpowered study wastes resources. If you can only recruit 50 patients,
you can only detect $d \geq 0.57$. Know this *before* you start collecting data.

Part V: Permutation Tests

No distributional assumptions — let the data speak

If H_0 Is True, Labels Don't Matter

Under H_0: drug and placebo groups come from the *same* distribution.
So the group labels are **arbitrary** — shuffling them shouldn't change anything.

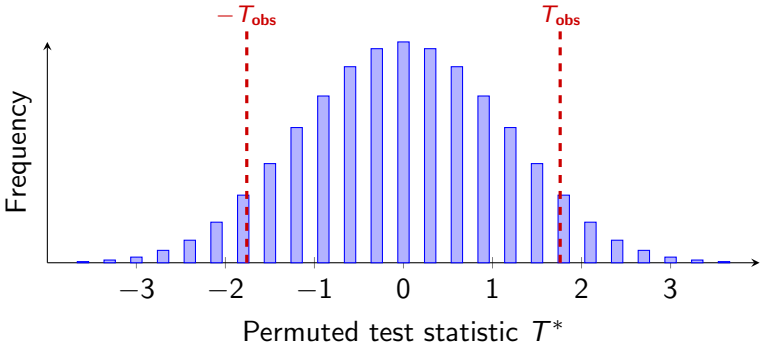


Algorithm:

1. Compute T_{obs} from the original data
2. Shuffle the group labels randomly (keeping group sizes fixed)
3. Compute T^* on the shuffled data
4. Repeat B times (e.g., $B = 10{,}000$)
5. p-value $= \frac{\#\{|T^*| \geq |T_{\text{obs}}|\}+1}{B+1}$

Permutation Distribution

Drug trial: $T_{\text{obs}} = -1.76$, $B = 10{,}000$ permutations



Count: ≈ 780 of 10,000 permutations have $|T^*| \geq 1.76$.
Permutation p-value: ≈ 0.078 — nearly identical to the z-test $p = 0.078$.

Permutation Tests: Pros and Cons

Advantages

+ No distributional assumptions
+ Exact p-values (not approximate)
+ Works for *any* test statistic (median, ratio, custom metric)
+ Easy to understand and implement

Limitations

– Computationally intensive (but cheap on modern machines)
– Tests the "sharp null" only (H_0: identical distributions)
– Cannot easily give CIs (use bootstrap for that)
– Needs exchangeability under H_0

Part VI: Multiple Testing

Test 20 hypotheses, and one will be "significant" by chance

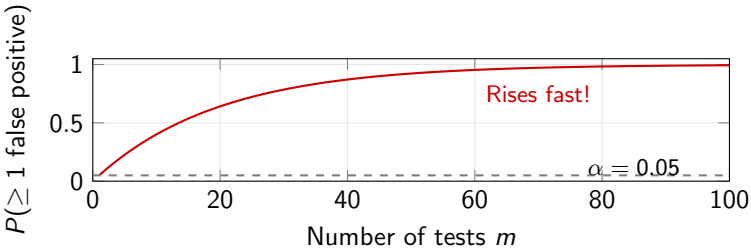
The Multiple Testing Problem

Examples: 20,000 genes, 50 A/B test variants, multiple clinical endpoints

If you test $m = 20$ independent true nulls at $\alpha = 0.05$:

$$P(\text{at least one false positive}) = 1 - (1 - \alpha)^m = 1 - 0.95^{20} = \mathbf{0.64}$$

64% chance of a false discovery — even when nothing is real!



Correcting for Multiple Tests

Bonferroni (FWER)

Reject if $p_i \leq \alpha/m$

For $m = 20$: threshold $= 0.0025$

Controls: $P(\text{any false positive}) \leq \alpha$

+ Simple and safe
– Very conservative when m is large

Benjamini–Hochberg (FDR)

Sort p-values: $p_{(1)} \leq \cdots \leq p_{(m)}$
Find largest k: $p_{(k)} \leq \frac{k}{m}\alpha$
Reject $H_{(1)}, \ldots, H_{(k)}$

Controls: expected *fraction* of false discoveries $\leq \alpha$

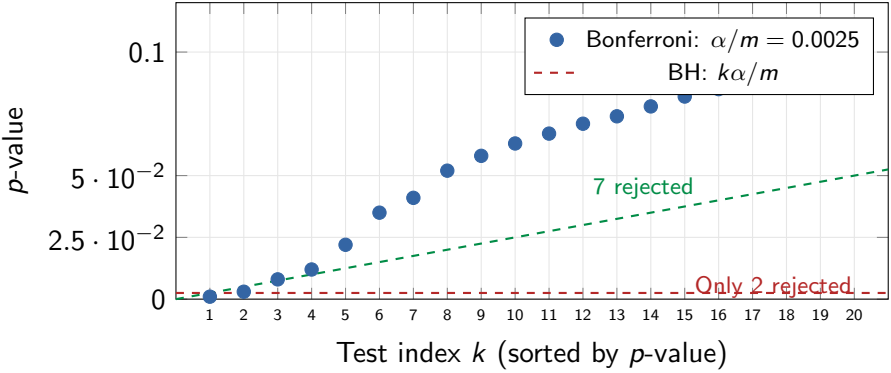
+ More powerful — the modern standard

FWER (Bonferroni): "Every single rejection must be real." Use for small m.

FDR (BH): "Some false discoveries OK if I find more true ones." Use for large m.

BH in Action: The Step-Up Procedure

$m = 20$ tests, $\alpha = 0.05$



Dots below a line = rejected by that method.
BH rejects 7 hypotheses; Bonferroni only 2. BH finds more while controlling the *fraction* of false discoveries.

Summary: Hypothesis Testing

Logic: Assume H_0 (nothing happening). Ask: how surprising is my data under H_0?

Test statistic: Compresses data into one number measuring distance from H_0.

p-value: $P(\text{this extreme or more} \mid H_0)$. **NOT** $P(H_0 \mid \text{data})$.

Type I (α): False alarm. **Type II (β):** Missed detection. **Power** $= 1 - \beta$.

Effect size: Cohen's d measures "how big?" independently of sample size.

Power analysis: Plan n before collecting data. Underpowered studies waste resources.

Permutation tests: Shuffle labels under H_0. No distributional assumptions needed.

Multiple testing: Many tests $\Rightarrow$ many false positives. Bonferroni (strict) or BH-FDR (modern).

Practical: Hypothesis Testing & Power

1. **z-test by hand:**
   - Drug trial: $n = 100$, $\bar{X}_D = 137$, $\bar{X}_P = 140$, $S = 12$
   - Compute T, find p-value, decide at $\alpha = 0.05$
2. **Power simulation:**
   - Under H_1: $\delta = 3$, $\sigma = 12$. Simulate 10,000 experiments
   - Try $n = 100$, 250, 500. What fraction reject H_0?
3. **Permutation test:**
   - With the drug trial data, implement a permutation test ($B = 10{,}000$)
   - Compare p_{perm} with $p_{\text{z-test}}$
4. **Multiple testing:**
   - Simulate 1,000 null experiments ($\mu_1 = \mu_2$). How many give $p < 0.05$?
   - Apply Bonferroni and BH. How many survive each correction?

Homework

1. A coffee shop claims cups contain 350 mL. You measure $n = 25$: $\bar{X} = 345$, $S = 10$.
   Test $H_0 : \mu = 350$ vs $H_1 : \mu \neq 350$ at $\alpha = 0.05$.
   Compute T, p-value, and 95% CI. Does the CI agree with the test?

2. A researcher wants to detect $d = 0.3$ with 80% power at $\alpha = 0.05$.
   (a) How large a sample per group?
   (b) If budget allows only $n = 50$ per group, what is the minimum detectable d?

3. You test 100 genes for disease association. 7 give $p < 0.05$.
   (a) How many false positives do you expect if all nulls are true?
   (b) Apply Bonferroni: how many survive?
   (c) Apply BH at $q = 0.10$: how many survive?

4. Two teaching methods. A scores: 78, 85, 91, 72, 88. B scores: 65, 70, 82, 68, 74.
   Implement a permutation test for the difference in means ($B = 10{,}000$).
   Is the difference significant at $\alpha = 0.05$?

Recommended Visualizations & Resources

Interactive: Seeing Theory — Frequentist Inference (Brown)

`seeing-theory.brown.edu/frequentist-inference` — drag sliders for rejection regions, Type I/II errors, power.

Interactive: R Psychologist — Understanding Statistical Power

`rpsychologist.com/d3/nhst/` — the best interactive power visualization. Drag effect size and n.

Video: StatQuest — p-Values, Clearly Explained

The clearest short explanation of what p-values are and are not. Also: "Hypothesis Testing and p-Values."

Reading: ASA Statement on p-Values (2016)

Six principles every scientist should know. `doi.org/10.1080/00031305.2016.1154108`

Reading: Wasserman, "All of Statistics" — Ch. 10

Hypothesis testing, Neyman–Pearson, power. Concise and rigorous.

Questions?

Next: Lecture 10 — t-tests, chi-squared, likelihood ratio tests